FINAL EXAM

Fall 2018

SOLUTIONS ARE AT THE END OF THIS FILE

- This is a closed-book exam.
- Students whose native language is not English may use a dictionary if they wish.
- Please check that you have 7 pages in your copy of the exam.
- There are 180 points in the exam. You also have 180 minutes.
- **Please put the first section, covering the shorter questions, in one book and the longer questions in a second book. Label all books**.
- Remember to put your name on all books.
- Do not be concerned if your answers are not whole numbers.
- You cannot use your phone as a calculator. Simple calculators will be provided by AASL office, and are sufficient for completing all calculations in the exam.
- If your answer is a number put a box around it. Example: $\boxed{\text{P=\$436, Q=35,465 .}}$

Section I: Short Questions [70 points]

Please be concise. None of the answers require more than a brief paragraph.

1) [10 points] The Government of Malawi is planning to subsidize families that choose to take flu shots. Do you think this intervention would be beneficial or detrimental to society? Please provide a precise economic argument for your view.

2) [10 points] A study by the Haas Avocado Board notes that that there would likely be a 10% decrease in avocado crop because of an unusual heatwave in early July in California. The study further predicts that this 10% decrease in supply would drive avocado prices in the US up by as much as 30%, a remarkably large impact. From reading this study, what can you infer about the elasticity of demand for avocado - do you think demand for avocado is likely to be elastic or inelastic? Does this make sense intuitively? Why or why not?

3) [10 points] A branded drug is a pharmaceutical that is covered by a patent. By law, only one firm can produce it, for a given length of time. After that time patents expire, and when this happens, a generic drug can enter the marketplace. The generic drug is chemically identical and is typically priced well below the price of the branded drug. It is common to observe that when a generic drug is introduced into the marketplace, the price of the branded (e.g. previously patented) drug increases. Explain why this might happen

4) [10 points] A local company assembles machines using domestically-sourced steel and imported precision aluminum. They had already solved for an optimal mix of these inputs when the US dollar began to appreciate. As the US dollar strengthens, would you expect them to change their input mix towards using more steel, more aluminum, or continue with the same mix? Provide a formal economic reason for your answer,

5) [10 points] A television pundit claimed that while it is true that an import tariff is likely to increase prices to consumers, since the government collects that tariff revenue, the country is not any worse off. Do you agree or not, and why?

6) [10 points] Many executive compensation packages include “stock options” as part of the compensation scheme. What is the economic rationale for this type of compensation, as opposed to, say, granting shares or paying a fixed salary?

7) [10 points] Philadelphia, the nation's largest city with a soda tax levies a tax of 1.5-cents-per-ounce on the manufacturers of sweetened beverages. Analysts predict that the price of a 12 pack of 12-ounce sodas would increase by $2.16. Do you agree with this prediction? Argue why or why not.

Section II: Long Questions – Total of 110 points

1. [20 points] Local ready-mix concrete market

Transition-Mix has a local monopoly over the ready-mix concrete industry in lower Manhattan. The demand for ready-mix concrete is given by $Q = 100-2P$, where Q is the quantity of ready-mix concrete in thousand cubic yards per month, and P is the price in dollars per cubic yard. Transition-Mix incurs a fixed cost of production of $12 and a variable cost of $5 per cubic yard.

a) [3 points] What are the profit maximizing price and quantity choices for Transition-Mix?

b) [3 points] What is the elasticity of demand at this price? Does this elasticity make sense to you? Why or why not?

c) [2 points] What is the total profit of Transition-Mix at the optimal price and quantity choice?

d) [4 points] If the market were instead perfectly competitive (with many ready-mix concrete providers (all with identical marginal costs as the monopolist), what would you predict to the price of ready-mix concrete? What would be the quantity sold?

e) [4 points] Let's go back to the setting with Transition-Mix being a monopoly (i.e., part (d) no longer applies.) The NY City government believes that local builders are suffering because of Transition-Mix's monopoly power and consequent high prices. To extract some surplus from the monopoly, the Government levies a \$1 tax (to be paid by Transition-Mix) for every cubic yard of concrete mix sold. Compute the new price and quantity that you expect in the market.

f) [4 points] Do you think the above tax is effective in helping the local builders? Explain **quantitatively** why or why not?

2. [15 points] Market for Raw Cookie Dough

Congratulations, you discovered a hot new trend: New York citizens are willing to pay \$5 for a serving of raw cookie dough. The market for cookie dough is perfectly competitive. The weekly total cost function for a typical firm that can sell raw cookie dough, as a function of the number of servings (measured by q), is:

$$TC(q) = 225 + q + 0.01q^2$$

Fixed costs represent the costs of renting in trendy areas, and are due **at the beginning of every week**.

a) [6 points] Solve for your firm's marginal cost, average total cost and supply functions (supply function should be in the form of q=…)

b) [4 points] What is your firm's output and profit in the short-run?

c) [5 points] It turns out that Cookie Dough is not much of a "differentiated" product. If there were truly free entry and no constraint on the number of firms that could enter, what would we expect the long-run price of raw cookie dough in New York to become?

3. [20 points] Price Discrimination at Pear Inc

The high-tech company Pear Inc. has developed an exciting new product: Myphone with a large interactive screen and many customizable features. Pear Inc. is considering the introduction of both a regular version which costs $200 to produce and a slightly higher-end version with more features called "Myphone Super" which costs $350 to produce. Pear's marketing departments suggests that consumers can be categorized as loyalists and casual users. Each consumer would consider at most one phone.

Consumer Type	Number of consumers	Willingness to pay for **Myphone**	Willingness to pay for **Myphone Super**
Casual	10 million	500	600
Loyalist	1 million	800	1100

You have been hired to advise Pear on its pricing strategy.

a) [15 points] Would you recommend that Pear Inc sell one particular model or both? What is the optimal pricing strategy that you would recommend to Pear Inc.? Consider all possible strategies and explain your reasoning systematically and completely.

b) [5 points] How would your answer change, if at all, if the causal segment were willing to pay $675 for the Myphone Super?

4. [15 points] Vehicle Insurance.

Vehicle insurance markets are a classic example of adverse selection. One feature of most states is a requirement to have vehicle insurance. Did you know that in New Hampshire, you are not required to carry insurance as a driver? Let us examine how the market performs in New Hampshire. The state has considered mandating insurance. We will examine the market for this vehicle insurance for drivers in the 30-35 age group.

In this age group, there are two types of drivers: “Safe” types, who make up 80% of this population, and “Risky” types, who make up the other 20%. The Safe types have expected annual accident expenses that are uniformly distributed between \$0 and \$1000. The Risky types are guaranteed to have \$1600 in annual expenses. All customers know their expected annual accident expenses, but insurers do not.

Math reminder: the mean of the uniform distribution between a and b is (a+b)/2, and the probability that a draw from such a distribution is at or above p is (b-p)/(b-a).

a) [5 points] First, suppose hypothetically that all individuals in this market (drivers between 30 and 35 in age) will purchase insurance plans because they are required to. We will assume the market is competitive, and that insurers have no cost advantages, so that premiums equal average costs. What will be the resulting policy premiums with all individuals covered?

b) [5 points] In reality, consumers will purchase an insurance plan only if the premium is less than their expected expenses (i.e. they are risk-neutral; making them risk-averse would complicate the math without changing the intuition). At the price you just found, what proportion of the Safe types would choose to purchase vehicle insurance if the requirement to purchase were lifted?

c) [5 points] The New Hampshire state government has proposed a “mandate,” which is a penalty that uninsured drivers must pay. What is the smallest penalty that will cause the market to arrive at the situation described in Part (a), where all consumers are purchasing the insurance.

5. [20 points] A Voting Game

A three-person committee has to choose a winner for a national painting prize. After some debate, there are three paintings still under consideration: a Cubist painting, an Expressionist painting, and a slightly more traditional Impressionist painting. Let’s call these paintings C, E and I; and call the committee members 1, 2 and 3. The preferences of the committee members are as follows:

1: C>E>I

2: I>C>E

3: E>I>C

For example, Member 1 strictly prefers the C to E and strictly prefers E to I.

The rules of the competition say that, if they disagree, they should vote (secret ballot, one member one vote, and the painting with the most votes wins). If the vote is tied (which happens when each painting has one vote), the winner will be the painting for which member 1 voted. So, it seems that member 1 might have an advantage. Consider this simultaneous voting game where each member has three strategies: C, E, or I, and answer the following questions:

[Hint: You do NOT need to know the exact payoffs. Any payoffs will be fine provided that they are consistent with the preference orders given above. And you do NOT need to draw a matrix form. Recall the definition of Nash equilibrium (NE) before you start.]

a) [6 points] Is it a Nash equilibrium that each member votes for their favorite painting (i.e. (C, I, E))? Please explain your answer.

b) [6 points] Is it possible that in a NE **member 1's least favorite painting** (i.e., painting I) wins but nobody votes for their least favorite painting (which is anyway a weakly dominated strategy)? If yes, please write down the NE which specifies each member's vote; if no, please explain.

c) [4 points] This game has multiple NE. You don't need to figure out all possible NE. But, among the following possible voting outcomes:
(C, E, I); (C, C, E); (E, I, I),
which one(s) is Nash equilibrium?

Now suppose the three members vote in a **sequential and public** way with the order of 1, 2 and 3, so that member 2 can observe member 1's vote and member 3 can observe both 1 and 2's votes.

d) [4 points] Which painting will then win the prize? (You are not required to draw the game-tree.)

6. [20 points] A Bargaining Game

Consider two players A and B. They need to divide a prize $50. The game goes as follows:

In the first round, player A proposes a division plan: x for herself and 50-x for player B. After receiving the proposal, player B decides whether to accept it or not. If it gets accepted, they divide the prize accordingly and the game ends. If it gets rejected, the game will continue to the second round, but the prize will be reduced to $40.

In the <u>second</u> round, it is player B's turn to make the proposal: y for himself and 40-y for player A. Then player A decides whether to accept it or not. If it gets accepted, they divide the prize accordingly and the game ends. If it gets rejected, the game ends and <u>both players get nothing</u>.

Suppose each player just wants to get as much money as possible for themselves, and **<u>they DO NOT care about anything else (e.g., fairness or the other player's welfare)</u>**.

A few extra rules (which you should read carefully):

(i) Consider integer numbers only.
(ii) Zero offer is not allowed (i.e., $1 \leq x \leq 49$, $1 \leq y \leq 39$).
(iii) When a player is indifferent between accepting and rejecting, let us suppose she or he will accept.

a) [5 points] If the game continues to the second round, how much will player B propose for himself?

b) [5 point] Anticipating that, how much will player A propose for herself in the first round?

Suppose now that the game potentially has three rounds. If player B's offer in the second round is rejected, the game will continue to the third round and the prize will be further reduced to \$30. In the third round, it is player A's turn again to make the proposal: z for herself and 30-z for player B, where $1 \leq z \leq 29$. Then player B decides whether to accept it or not. If it gets accepted, they divide the prize accordingly; otherwise, the game ends and both players get nothing.

c) [5 points] How much will player A then propose for herself in the first round?

Suppose now there is an **additional "fairness" constraint** in any round of the bargaining game: no player accepts an offer which gives her/him \$6 or more less than the other player.

d) [5 points] How will your answer to question (c) change?

SOLUTIONS

Section I: Short Questions [70 points]

Please be concise. None of the answers require more than a brief paragraph.

1) Vaccines have positive externalities – this means people underestimate the benefit of a vaccine and therefore tend to under-consume. A subsidy can help alleviate this distortion.

2) If a relatively small shift in the supply curve causes a big change in prices, this indicates that the demand curve must be quite steep. So you would expect demand for avocado to be inelastic.

This can make sense. Possible reasons: (i) The amount a household spends on avocado is likely very small as a proportion of total budget and so you can expect that people aren't very price sensitive. (ii) Very important ingredient for many restaurants because there aren't good substitutes (because it is a very popular item) – so they are less price sensitive.

3) The availability of cheaper substitutes will make some consumers move from using the branded product to using the generic. The consumers who continue to buy the branded drug are the less price elastic consumers. The branded firm now faces a customer segment that is less price sensitive. This means the firm with the branded drug can raise prices.

4) A stronger US dollar means imported aluminum is cheaper; they would move towards using more aluminum. The MPI of aluminum would increase, but MPI of steel stays the same.

5) The tariff introduces deadweight loss; the government revenue is less than the surplus destroyed. Therefore, no, the country is indeed worse off. If they mention DWL, full credit.

6) The issue is that stock options are all "upside" and no downside. These "convexify" the CEO's utility function. These "correct" a risk-averse CEO's incentives around the share price (make CEOs behave in the risk-neutral way that shareholders want them to, while pure share grants or salary don't address risk aversion).

7) No, the price would not go up by \$2.16 (the full extent of the tax). This is because the incidence of tax is shared by manufacturers and consumers based on the relative elasticity of demand and supply. So tax will be partly borne by manufacturers too and price will go up by less than \$2.16

Section II: Long Questions – Total of 110 points

1. [20 points] Local ready-mix concrete market

a)

P=50-0.5Q
Revenue=50Q-0.5Q^2
MR=50-Q
So MR=MC gives 50-Q=5 or Q=45 and P=27.5

b)

Elasticity = -2*(27.5/45)=-1.2

c)

Profit = (27.5-5)*45-12=1000.5

d)

For competitive market, you can think of supply curve as MC curve at P=5.
So market equilibrium would be 50-0.5Q=5, which gives Q=90, P=5.

e)

The tax will be like increasing the per unit cost of the monopolist by 1.
To find profit maximizing price and quantity, we now have
50-Q=6 which gives Q=44 and P = 28.

f)

Not effective.
Price is now higher for consumer and less quantity being sold. So consumers worse off.
Earlier, the CS of the local builders was 0.5*(50-27.5)*45=506.25
New CS = 0.5*(50-28)*44 =484
So CS went down by 22.25

2. [15 points] Market for Raw Cookie Dough

a)

To get the marginal cost, take the derivative of total cost with respect to q:

$$mc = \frac{\partial TC}{\partial q} = 1 + .02q$$

Average total cost is given by:

$$ac = \frac{TC}{q} = \frac{225}{q} + 1 + .01q$$

As the firms take the price as given, the supply function can be found by equating price and marginal cost, then solving for q:

$$p = 1 + 0.02q_s$$

$$\Longrightarrow q_s = 50(p - 1)$$

b)

In the short run, the current prevailing wholesale market price is $5 per serving.

Plug in the market price into the supply function, yielding:

$$q = 200$$

With this, we can solve for a firm's total profit:

$$\pi = TR - TC$$

$$= (200 * 5) - 225 - 1(200) - 0.01(200^2) = \$175$$

c)
Approach 1: We know that in the presence of free entry that in the long-run firms will make zero economic profit. This implies that the long-run price of dough will be the price such that the average total cost is at its minimum.

$$\frac{\partial ATC}{\partial q} = -\frac{225}{q^2} + .01$$

Setting this equation equal to zero and solving for q, yields:

$$q^2 = 22500 \Rightarrow q = 150$$

The long-run price can then be found by evaluating the average total cost function at q = 150:

$$ATC(150) = 4$$

Approach 2: You could set MC=AC to get min AC level.

$$1 + .02q = \frac{225}{q} + 1 + .01q$$
$$.02q = \frac{225}{q} + .01q$$
$$.01q^2 = 225$$

which would give

$$q^2 = 22500$$

The long-run price can then be found by plugging in q = 150 in ATC function:

$$ATC(150) = 4$$

3. [20 points] Price Discrimination at Pear Inc

a)

Strategy 2: Pear Inc. can choose to sell only Myphone.

Then should sell at $500. This gives profit of (500-200)x11 million= 3.3 billion.

Strategy 1: Pear Inc. can choose to sell only Myphone Super

Then, should sell at $600. This gives profit of (600-350)x11 million= 2.75 billion.

Strategy 3: Pear Inc. can choose to sell both products.

In this case, optimal strategy will be to sell Myphone at $500 and Myphone Super at 800.

(To get the price of 800, notice that we need to set price of Myphone Super such that 1100-p>=800-500, i.e. p<=800.)

This gives profit of (500-200)x10 million+(800-350)x1million= 3.45 billion.

So this is the best.

b)

Notice nothing changes in Strategy 2 & 3 above.

But now Strategy 1 gives a higher profit. So Pear Inc should just sell Myphone Super at $675. This gives profit of (675-350)x11 million= 3.575 billion.

4. [15 points] Vehicle Insurance.

a)

Solution:

Policy premiums = ______$720________

First note that the average expense of a safe driver is $\frac{\$1000+\$0}{2}$ = \$500. The average cost is therefore:

$$AC(p) = 80\% \times \$500 + 20\% \times \$1600 = \$720$$

In a competitive market, the premiums will equal the average cost of \$720.

b)

Solution:

% of Safe Types = _______28 %_________

We know that a consumer will buy the policy if their expected expense is greater than the premium. For safe drivers, the proportion that will buy the policy at \$320 is:

$$= \frac{\$1000 - \$720}{\$1000} = 0.28$$

So only 28% of safe drivers will buy the plan when the premium is \$720.

Right away, we see the challenge without universal coverage: without some way of forcing individuals to buy insurance, the safest drivers in a market would choose not to participate, which would increase the average cost, which would increase prices, and so on.

c)

Solution:

Penalty = ________$720___________

This penalty must be set such that safe drivers with no expenses will buy the plan. This is $720.

5. [20 points] A Voting Game

a)
No.
(C, I, E) is not a NE because member 3 would have a unilateral incentive to switch to I so that her/his second favorite painting wins. (No other players want to deviate unilaterally.)

b)

(C, I, I)

c)

(C, C, E)
[In the first one, member 2 or 3 wants to deviate; in the third one, member 3 wants to deviate. This is not required for the answer.]

d)
If member 1 votes for C, then member 2 will vote for I and member 3 will vote for I as well. So I will win, but that's the worst outcome for member 1.
If member 1 votes for E, then no matter what member 2 will vote, member 3 will vote for E. Therefore, E will win.
If member 2 votes for I, I will win as in the first case.
As a result, in this sequential voting game, painting E will win.

6. [20 points] A Bargaining Game

a)

Since this is the last round, Player B will propose **39** for himself.

b)
Since Player B will reject any offer below 39, Player A will propose **11** for herself (and so 39 for Player B in which case Player B will just accept).

c)

In the last round (if reached), player A will propose 29 for herself.

Anticipating that, in the second round (if reached), player B will propose 11 for himself (and so 29 for player A so that player A will accept).

Anticipating that, in the first round, player A will propose **39** for herself (and so 11 for player B so that player B will accept).

d)

In the last round (if reached), player A will propose 17 for herself (and so 13 for B so that the difference is less than 6 and B will accept).

Anticipating that, in the second round (if reached), player B will propose 22 for himself (and so 18 for player A so that player A will accept). (Notice the difference is less than 6. If B offers 23 to himself and 17 to A, the difference will be 6 and so the offer will be rejected.)

Anticipating that, in the first round, player A will propose **27** for herself (and so 23 for player B so that player B will accept). (Notice that the difference is less than 6 so satisfies the fairness constraint. 28 and 22 don't work here.)